

34th Nordic Mathematical Contest 2020

Solutions

Problem 1 For a positive integer n , denote by $g(n)$ the number of strictly ascending triples chosen from the set $\{1, 2, \dots, n\}$. Find the least positive integer n such that the following holds: The number $g(n)$ can be written as the product of three different prime numbers which form an arithmetic progression with common difference 336.

Solution to Problem 1 The number of such sequences is $\binom{n}{3}$, since the sequence is uniquely determined by the three numbers it contains.

If $\binom{n}{3}$ is the product of three distinct prime numbers, then n must be odd (otherwise $n(n-1)(n-2)$ contains too many factors of 2). We have three different possibilities (where p_i are distinct primes):

- Case 1: $n = p_1$, $n - 1 = 6p_2$, and $n - 2 = p_3$
- Case 2a: $n = 3p_1$, $n - 1 = 2p_2$, and $n - 2 = p_3$
- Case 2b: $n = p_1$, $n - 1 = 2p_2$, and $n - 2 = 3p_3$

In Case 1, $p_1 - p_3 = 2$, so these numbers cannot form an arithmetic progression with a common difference of 336.

In Case 2a, we get:

$$p_1 = \frac{n}{3}, \quad p_2 = \frac{n-1}{2}, \quad p_3 = n-2$$

The smallest difference here is:

$$p_2 - p_1 = \frac{n-1}{2} - \frac{n}{3} = \frac{n-3}{6}$$

We optimistically hope this step is $1 \cdot 336$, which is the theoretical minimum. Then:

$$6 \cdot 336 = n - 3 \Rightarrow n = 2019$$

It turns out the next step to $n - 2$ is $4 \cdot 336$. This suggests that this is the correct answer, and it is worth checking whether:

$$\frac{2019}{3} = 673, \quad \frac{2018}{2} = 1009, \quad \text{and } 2017$$

are prime numbers. This requires some effort, especially checking 2017, which involves testing against primes up to 43.

We still need to check whether Case 2b can yield a smaller n . Here, the smallest difference is:

$$p_2 - p_3 = \frac{n-1}{2} - \frac{n-2}{3} = \frac{n+1}{6}$$

Again, we hope this equals $1 \cdot 336$, which gives:

$$6 \cdot 336 = n + 1 \Rightarrow n = 2015$$

But 2015 is divisible by 5, which excludes the possibility that $\binom{2015}{3}$ can be written as a product of exactly three prime numbers.

Hence the answer is $n = 2019$.

Problem 2 Georg has $2n + 1$ cards with one number written on each card. On one card the integer 0 is written, and among the rest of the cards, the integers $k = 1, \dots, n$ appear, each twice. Georg wants to place the cards in a row in such a way that the 0-card is in the middle, and for each $k = 1, \dots, n$, the two cards with the number k have the distance k (meaning that there are exactly $k - 1$ cards between them). For which $1 \leq n \leq 10$ is this possible?

Solution to Problem 2 Let the positions be numbered from $-n$ to n , and let a_k and $a_k - k$ be the positions of the two cards with the number k .

By symmetry around 0, we get:

$$0 = \sum_{k=1}^n (a_k + a_k - k) = 2 \sum_{k=1}^n a_k - \frac{n(n+1)}{2}$$

It follows that either n or $n + 1$ must be divisible by 4, i.e., $n \in \{3, 4, 7, 8\}$.

The following configurations show that each of these values is indeed possible:

- For $n = 3$: 2 3 2 0 3 1 1
- For $n = 4$: 2 3 0 4 3 1 1 4
- For $n = 7$: 5 3 4 7 3 5 4 0 6 2 7 2 1 1 6
- For $n = 8$: 6 8 5 7 1 1 6 5 0 8 7 4 2 3 2 4 3

Problem 3 Each of the sides AB and CD of a convex quadrilateral $ABCD$ is divided into three equal parts, $|AE| = |EF| = |FB|$, $|DP| = |PQ| = |QC|$. The diagonals of $AEPD$ and $FBCQ$ intersect at M and N , respectively. Prove that the sum of the areas of $\triangle AMD$ and $\triangle BNC$ is equal to the sum of the areas of $\triangle EPM$ and $\triangle FNQ$.

Solution to Problem 3

Let $T(XYZ)$ denote the area of triangle XYZ (similarly for quadrilaterals).

We first note the following equalities of areas:

$$T(EEP) = T(AEP), \quad T(PFQ) = T(QFC), \quad T(EFQ) = T(FBQ), \quad T(EQP) = T(EPD)$$

From the first two lines, we get:

$$T(EFQP) = T(AEP) + T(QFC)$$

From the third and fourth lines:

$$T(EFQP) = T(FBQ) + T(EDP)$$

Adding both expressions for $T(EFQP)$ gives:

$$2T(EFQP) = T(AEPD) + T(EPM) - T(AMD) + T(FBCQ) + T(FNQ) - T(BNC)$$

This means the problem is equivalent to showing:

$$2T(EFQP) = T(AEPD) + T(FBCQ)$$

The area of a quadrilateral is half the magnitude of the cross product of its diagonals.

Let $\vec{t} = \vec{EA} + \vec{DP}$. Then we can write the three cross products as:

$$2T(AEPD) = \vec{AP} \times \vec{ED}$$

$$2T(EFQP) = \vec{EQ} \times \vec{FP} = (\vec{AP} + \vec{t}) \times (\vec{ED} + \vec{t})$$

$$= \vec{AP} \times \vec{ED} + \vec{t} \times \vec{ED} + \vec{AP} \times \vec{t}$$

$$2T(FBCQ) = \vec{FC} \times \vec{BQ} = (\vec{AP} + 2\vec{t}) \times (\vec{ED} + 2\vec{t})$$

$$= \vec{AP} \times \vec{ED} + 2\vec{t} \times \vec{ED} + 2\vec{AP} \times \vec{t}$$

From this, the result follows.

Problem 4 Find all functions $f : \mathbb{R} \setminus \{-1\} \rightarrow \mathbb{R}$ such that

$$f(x)f\left(f\left(\frac{1-y}{1+y}\right)\right) = f\left(\frac{x+y}{xy+1}\right)$$

for all $x, y \in \mathbb{R}$ that satisfy $(x+1)(y+1)(xy+1) \neq 0$.

Solution to Problem 4

We are given the functional equation:

$$f(x)f\left(f\left(\frac{1-y}{1+y}\right)\right) = f\left(\frac{x+y}{xy+1}\right)$$

for all $x, y \in \mathbb{R}$ such that $(x+1)(y+1)(xy+1) \neq 0$.

Set $y = 1$:

$$f(x)f(f(0)) = f(1)$$

If $f(f(0)) \neq 0$, we can divide and obtain $f(x) = c$ for some constant $c \in \mathbb{R}$. The only such constant solutions are $f(x) = 0$ and $f(x) = 1$.

So we assume $f(f(0)) = f(1) = 0$.

Set $x = y = 0$:

$$f(0)f(f(1)) = f(0)^2 = f(0) \Rightarrow f(0) = 0 \text{ or } 1$$

If $f(0) = 0$, then setting $x = 0$ gives $f(y) = 0$ for all y , which contradicts our assumption. So we assume $f(0) = 1$.

Then:

$$f\left(f\left(\frac{1-y}{1+y}\right)\right) = f(y)$$

Substitute this back into the original equation:

$$f(x)f(y) = f\left(\frac{x+y}{xy+1}\right)$$

Let $z = \frac{1-y}{1+y}$, then:

$$f(f(z)) = f\left(\frac{1-z}{1+z}\right)$$

We now prove the lemma that:

$$f(x) = 0 \iff x = 1$$

We know $f(1) = 0$, so assume $f(a) = 0$ for some $a \neq 1$. Then by the identity

$$f(x)f(a) = f\left(\frac{x+a}{xa+1}\right),$$

we get $f\left(\frac{x+a}{xa+1}\right) = 0$ for all $x \in \mathbb{R} \setminus \{-1, -1/a\}$. Since $z = \frac{x+a}{xa+1}$ can take any value in $\mathbb{R} \setminus \{-1, 1/a\}$, this implies $f(z) = 0$ for all z , contradicting our assumption. Hence, $f(x) = 0 \iff x = 1$.

Next, we show:

$$f(a) = f(b) \Rightarrow a = b \text{ or } a = \frac{1}{b}$$

Assume $a \neq b$ and $f(a) = f(b)$. Since $f(b) \neq 0$, we use the identity again:

$$f(b)f(-b) = f(0) = 1 \Rightarrow f(-b) = \frac{1}{f(b)}$$

Then:

$$f(f(a)f(-b)) = f\left(\frac{1-ab}{1+ab}\right)$$

Using the lemma, this implies:

$$\frac{1-ab}{1+ab} = 1 \Rightarrow ab = 0$$

which contradicts the domain. Therefore, $f(a) = f(b) \Rightarrow a = b$ or $a = \frac{1}{b}$.

Combining this with the earlier result, we conclude that for each $x \in \mathbb{R} \setminus \{-1\}$, the only possible solutions are:

$$f(x) = \frac{1-x}{1+x}$$

This completes the solution.